

Presentation Script — AI in FinTech: Credit Scorecard System

Format: Live demo on the **RiskLens** dashboard (not a PowerPoint).

Total length: ~25–30 minutes · **6 speakers** · ~4–5 min each.

Backed by: `Final/fintech (1).pdf` (Tables 13 / 15 / 16) and `matlab/FinTech_Scorecard_Project.m`.

Before going live:

`cd fintech-dashboard && npm run dev` → open the dashboard, log in if needed, sit on **Slide 1 — Overview (/)**.

Keep the report PDF open on a second device as backup.

Use the numbered sidebar (1–11) to navigate — every speaker advances by clicking the next step.

Slide map (follow the numbered sidebar)

Step	Page	Speaker
1	Overview (/)	Speakers 1 & 2
2	WOE & Binning (/woe)	Speaker 3
3	Scorecard (/scorecard)	Speaker 3
4	ROC & Validation (/validation)	Speaker 4
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6	Accept / Reject (/decisions)	Speaker 5
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8	Risk Pricing (/pricing)	Speaker 5
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10	Final Decision (/final)	Speaker 6
11	Live Client Test (/test)	Speaker 6
12	MATLAB Charts (/outputs/matlab)	Optional appendix

Speaker 1 — Introduction & dashboard tour

Time: ~4 minutes · **Page:** Slide 1 — **Overview** (/)

[Stand near the screen. Dashboard already on Slide 1. Pause 2 seconds, eye contact, smile.]

Good morning everyone. We're presenting our **AI in FinTech** project — a complete credit risk system that takes a real loan portfolio, scores each applicant, decides who to accept, calculates the expected loss in dollars, and sets a fair interest rate for every approved client.

Instead of slides, we'll show you the entire project **live on our dashboard**, called **RiskLens**. Everything you see here is built on top of our MATLAB model and our final report — same numbers, same logic, just made interactive so you can see the story unfold.

[Point at the badge that says “Executive Dashboard” and read the title.]

This first page is the **Overview** — it gives the headline numbers before we dig in.

[Point at the four metric cards across the top, one by one.]

Four key facts about the project:

- **500 historical records** — clients with known outcomes used to *learn* who tends to default.
- **AUC of 0.6775** — a standard measure of how well the model separates good clients from bad. We'll explain this in detail later.
- **20 portfolio clients** — these are new loan applications waiting for a decision: **14 accepted, 6 rejected**.
- **\$1.4 million in accepted exposure** with an **expected loss of \$129,329.63** — that's the money at risk on the loans we'd actually book.

[Pause for 2 seconds. Emphasize this result.]

So at a glance: the dashboard answers a real bank question — *out of 20 people knocking on the door, who do we lend to, how much can we lose, and what should we charge?*

[Scroll slightly down to the “Key Parameters” panel on the right.]

On the right we list the **ground rules** used everywhere in the project. Every loan is **\$100,000**, the recovery rate if the client defaults is **60%**, so the **Loss Given Default — LGD — is 40%**. Our cutoff score for accepting a loan is **48.2032**, scores go from **0 to 100**, and the minimum interest rate formula is **PD × LGD**. We'll come back to all of these.

[Pause briefly. Look at the sidebar.]

One thing worth pointing out: notice the **numbered sidebar** on the left — steps **1 through 11**. That's the order we'll walk through today. Each of my teammates will take you through one or more of those steps live.

[Hover over Step 1 in the sidebar, then turn back to the audience.]

Now my colleague will explain the **business problem we're solving** and the **data behind the model**.

Transition → “To make sense of these numbers, let's start with the problem itself and the data we used.”

Speaker 2 — Business problem, data, and methodology overview

Time: ~4–5 minutes · **Page:** stays on Slide 1 — **Overview (/)**

[Take over the screen. Stay on Slide 1 — Overview. Look at the audience.]

Thank you. Before talking about charts and metrics, let's answer the obvious question: **why does any of this matter?**

When a bank receives a loan application, it really only has two ways to be wrong. If it accepts too many risky clients, it loses money on **defaults**. If it rejects too many good clients, it loses **revenue and market share**. The whole point of a **credit scorecard** is to rank applicants from safer to riskier using simple facts the bank already knows — **age, income, whether they own or rent, and whether they're employed**. A good model is one that puts most defaulters at the bottom and most good payers at the top.

[Scroll slightly down to the “Default Distribution — Historical Data” pie chart.]

This is our **training dataset: 500 historical clients** from the bank's records — read from the `HistoricalData` sheet of `DataProjScoreCard.xlsx`.

[Point at the pie chart and the numbers next to it.]

- **320 clients paid back normally** — coded as **0**, the green slice.
- **180 clients defaulted** — coded as **1**, the red slice.
- That's a portfolio default rate of **36%**.

[Emphasize this result.]

Thirty-six percent is **high** — much higher than what a real bank would tolerate without intervention. That's exactly why the bank needs a scorecard: a tool to push that number down by being more selective.

[Scroll down so the four “Default Rate by...” charts are visible.]

Now look at where the defaults are coming from. We split the 500 clients by **age, income, residential status, and employment** — and the story is very clear.

[Point at “Default Rate by Income Band”.]

Income is the strongest signal. Clients earning **under \$30k** default about **57%** of the time. Clients earning **over \$50k** default only **21%**. That's a huge gap.

[Point at “Default Rate by Age Group”.]

Age tells a similar story — under-35s default **50%** of the time, while clients over 65 only **17%**.

[Point at the residential and employment charts.]

Renters default more than **homeowners** — 40% vs 31%. **Unemployed or self-employed** ("Other") clients default at **44%** versus **28%** for fully employed clients.

[Pause. Look at the audience.]

These four patterns are exactly what the model is going to learn. We're not inventing a formula — we're letting the data show us **which combinations of characteristics correlate with repayment**.

[Scroll down to the “MATLAB Workflow Pipeline” strip — do NOT read every step word-for-word.]

Quickly, the pipeline behind everything you'll see today: we **load the data**, let MATLAB **group clients into risk buckets**, fit a **logistic regression**, **rescale the result into a 0–100 score**, **validate it** on the same historical data, and then **apply it to 20 new applicants** to make accept/reject decisions, calculate expected losses, and price the loans.

[Mention this business insight clearly.]

The important point: this is **AI in FinTech**, not a black-box machine-learning experiment. Every decision is traceable — we can always tell the bank exactly *why* a client got a particular score and exactly *why* a loan was rejected. That transparency is critical for credit risk, audit, and regulation.

[Pause for 2 seconds.]

So we have **500 labelled past clients** and **20 new applicants** to evaluate. Next, my colleague will explain how MATLAB turns these four features into a credit score.

Transition → “Let's see how MATLAB groups the data and builds the actual scorecard.”

Speaker 3 — Binning, WOE, Information Value & Scorecard construction

Time: ~4–5 minutes · **Pages:** Slide 2 — **WOE & Binning** (/woe) then Slide 3 — **Scorecard** (/scorecard)

[Click step 2 in the sidebar — WOE & Binning. Wait for the page to load.]

We're now on **Slide 2 — WOE & Binning Analysis**. This is where the model starts to take shape.

[Point at the page subtitle: “Autobinning on 500 historical clients”.]

In MATLAB, we use the **Credit Scorecard Toolbox** and call a function called **autobinning**. Its job is to take each predictor — age, income, employment, housing — and **split clients into groups** in the smartest possible way: groups where the default rate inside each group is very different from the next group. Income, for example, isn't used as a raw number; it's used as **"is this client under \$28k, between \$28–32k, \$32–35k, \$35–41k,"** and so on.

[Point at the four cards at the top of the page — IV summary.]

Once the groups are formed, MATLAB computes a number called **Information Value — IV** — for each predictor. IV measures how much **separating power** a predictor carries. The higher the IV, the more useful that variable is for predicting default.

Reading from the cards:

- **Income — IV 0.2355 — Strong** — our most powerful predictor.
- **Age — IV 0.1879 — Medium.**
- **EmploymentStatus — IV 0.1143 — Medium.**
- **ResidentialStatus — IV 0.0417 — Weak**, but still useful when combined.

[Pause for 2 seconds. Mention this business insight clearly.]

The bank now knows, *before* the model is even fitted, which questions matter most on a loan application form. Income matters more than housing status. That's already a useful business finding.

[Scroll to the main WOE chart — Income should be the default selected predictor.]

The big chart in the middle shows **Weight of Evidence — WOE — for income**. Each bar is one bin. **Green bars above the zero line** mean that bin is **safer than the portfolio average**. **Red bars below zero** mean **riskier than average**.

[Trace the bars from left to right.]

You can see it almost perfectly. Clients earning **under \$28k** sit deep in the red — WOE of about **–0.98**, the riskiest bin. Then risk decreases as income increases. By the time we reach **\$50k+** the

WOE is **+0.72** — clearly safer. That smooth shape — bad to good — is what the logistic regression will use.

[Click the “Age” tab above the chart.]

Same idea for **age**. Under 38 — red. Above 49 — strongly green, WOE around **+0.54**. Older clients in our data behave better.

[Click “EmploymentStatus” and then “ResidentialStatus”.]

For employment, **Employed** is green, **Other** is red — a clean split. For residential status, **HomeOwner** is green, **renter** is red, but smaller in magnitude — which matches its low IV.

[Scroll down to the four predictor cards with bin tables.]

Each predictor has a small table — bin, number of clients, percentage of defaults, and the WOE — exactly the values published in **Table 8 and 9 of our report**.

[Pause. Look at the audience.]

OK — so WOE turns categorical information into numbers the regression can use. Let's see what comes out the other end.

[Click step 3 in the sidebar — Scorecard.]

We're now on **Slide 3 — Credit Scorecard Points**.

[Point at the four charts on the page — one per predictor.]

After WOE, MATLAB fits a **logistic regression** on all four predictors using `fitmodel`, and then calls `formatpoints` to **rescale everything into a 0–100 score**. The four charts on this page show how many **points** each bin contributes.

[Point at the Income chart.]

Income, again, dominates: the **lowest** bin contributes **–8.68 points**, the **highest** contributes **+28.91 points**. That's a swing of almost **38 points** just from income alone.

[Point at the EmploymentStatus chart.]

Employment is also strong — **Employed** adds **+25.12 points**, **Other** only **+1.80**.

[Point at the ResidentialStatus chart.]

Housing adds **+21.10** for owners, **+6.71** for renters.

[Point at the Age chart.]

Age adds from **+0.17** for the youngest bin up to **+24.87** for the oldest.

[Emphasize this result.]

So a client's **final credit score** is just: their **age points + income points + housing points + employment points**, clamped to **0–100**. Higher score → lower risk. That's the whole scorecard, on one page, end to end.

[Pause for 2 seconds.]

The model is built. Now the question is: **does it actually work?** That's what my colleague will show next.

Transition → “Let's validate the model on the historical data and find the right cutoff.”

Speaker 4 — Validation (ROC, KS) + Portfolio scoring (PD)

Time: ~4–5 minutes · **Pages:** Slide 4 — **ROC & Validation** (/validation) then Slide 5 — **Portfolio Risk** (/portfolio)

[Click step 4 in the sidebar — ROC & Validation.]

We're now on **Slide 4 — ROC Curve & Validation Metrics**. This page answers the question my teammate just asked: **how good is the scorecard?**

[Point at the four metric cards at the top.]

We applied the function `validateModel` to the **same 500 historical clients** — and the dashboard shows the four official numbers from **Table 13 of our report**:

- **AUC = 0.6775**
- **KS Statistic = 0.2892**
- **KS Optimal Score = 48.2032**
- **Accuracy Ratio = 0.3551**

Let me unpack these one at a time.

[Point at the ROC curve chart on the left. Trace the teal curve with the cursor.]

The **ROC curve** plots, at every possible cutoff score, how many bad clients we'd catch versus how many good clients we'd wrongly reject. The **teal line is our model**. The **dashed grey diagonal is what you'd get from random guessing**.

[Pause. Emphasize this result.]

Our curve sits **clearly above the diagonal** across every cutoff. That means at any acceptance threshold, our model catches more defaulters than chance — the model has **real predictive power**.

[Point at the AUC card.]

AUC — Area Under the Curve — is the area between the teal line and the bottom axis. An AUC of **0.6775** means: if you pick one defaulter and one non-defaulter at random from the historical data, **there's about a 68% chance our model gives the defaulter the lower score**. For a four-variable model on 500 records, that's a solid, defensible result.

[Point at the KS card.]

KS — Kolmogorov–Smirnov — measures the **biggest gap** between the score distribution of good clients and bad clients. Our KS is **0.2892**, and crucially, it happens at the score value **48.2032** — that's the **KS Optimal Score** card.

[Point at the KS Optimal Score card.]

This number — **48.2032** — is the **single most important decision in the entire project**. It is the **acceptance threshold**. We're not picking it from intuition; the data itself tells us this is the score where good and bad clients are most cleanly separated.

[Read the rule slowly. Emphasize each word.]

Accept if the credit score is ≥ 48.2032 . Reject otherwise.

[Point at the “Validation Summary” panel on the right.]

The panel on the right is just a clean restate of the same numbers, plus a note that we deliberately train and validate on the same 500 historical clients — that's the methodology required by our project brief.

[Pause. Look at the audience.]

Now we have a working scorecard and a clear rule. Let's apply it.

[Click step 5 in the sidebar — Portfolio Risk.]

We're now on **Slide 5 — Portfolio Scoring**. This is the moment we move from past data to the **20 real applicants** the bank wants us to evaluate — the `ActualPortfolioData` sheet.

[Point at the scatter chart.]

Each dot is one of the 20 applicants. The **x-axis is their credit score**, the **y-axis is their Probability of Default — PD** — calculated by `probdefault` in MATLAB.

[Point at the red dashed vertical line at 48.2032.]

This red dashed line is the **cutoff at 48.2032** — exactly the KS optimal score we just saw. Everything to the right gets a loan, everything to the left is rejected.

[Point at the cloud of dots — green vs red.]

Green dots = Accepted, red dots = Rejected. And notice the shape — the dots fall **from upper-left to lower-right. Higher score → lower PD**. That monotonic relationship is exactly what you want from a credit scorecard: a higher score consistently means a safer client.

[Hover over a green dot near the top of the green cluster — Client 4 with score 100.]

The two safest clients have a perfect **score of 100** and a PD of just **12.79%**.

[Hover over the lowest red dot — Client 1.]

The riskiest applicant has a **score of 0** and a PD of **70.59%**. That's a client where 7 out of 10 simulations end in default — clearly not someone we want on the books.

[Scroll down to the “Risk band segmentation” cards.]

We also segment all 20 applicants into **three risk bands** for monitoring:

- **Low Risk** (score ≥ 80): **5 clients**
- **Medium Risk** (50–80): **9 clients**
- **High Risk** (below 50): **6 clients** — all six are the rejected ones.

[Pause for 2 seconds. Mention this business insight clearly.]

Just by sorting by score, we've already created **risk tiers** — the bank can now treat low-risk clients differently from medium-risk ones in terms of monitoring, communication, and pricing.

[Pause.]

We have scores. We have decisions. Now it's time to translate those into **dollars and cents**.

Transition → “Now let's turn these scores into actual business outcomes — who gets the loan, how much we could lose, and what we should charge them.”

Speaker 5 — Decisions, Expected Loss, Risk-Based Pricing

Time: ~4–5 minutes · **Pages:** Slide 6 — **Decisions** (/decisions), Slide 7 — **Expected Loss** (/expected-loss), Slide 8 — **Pricing** (/pricing)

[Click step 6 in the sidebar — Accept / Reject.]

We're now on **Slide 6 — Accepted vs Rejected Clients**. This page operationalizes the cutoff we just saw.

[Point at the “Decision Breakdown” bar chart on the left.]

The headline: out of 20 applicants — applying the rule **score $\geq$ 48.2032** — we get **14 accepted** and **6 rejected**. That's a **70% acceptance rate**.

[Point at the “Client Scores vs Threshold” bar chart on the right.]

This chart shows every client sorted by score. **Green bars are accepted, red bars are rejected**, and the **red dashed line at 48.20 is the cutoff**.

[Point to the rightmost green bars — clients 4 and 5.]

On the safe end you've got clients **4 and 5** with the maximum **score of 100**.

[Point at the borderline green bar — client 8 at ~55.84.]

This green bar here, hovering just above the cutoff, is **Client 8 at score 55.84**. He's a borderline case — we accept him, but he's our most risky accepted client.

[Point at the leftmost red bar.]

And at the bottom, **Client 1 with a score of 0** — clearly out.

[Scroll down to the “Portfolio results table”.]

Now the **full table** — this matches **Table 15 of our report**, one row per client.

[Point at the column headers.]

For every client we see their **ID, score, PD, decision, risk band**, and **expected loss in dollars**.

[Point at row for Client 4.]

Client 4: score **100**, PD **12.79%**, **Accepted, Low Risk**, expected loss **\$5,114** — that's the cheapest, safest client in the portfolio.

[Point at row for Client 1.]

Client 1: score **0**, PD **70.59%**, **Rejected**, **High Risk**, would have cost us **\$28,234** in expected loss — exactly why we don't lend to him.

[Point at row for Client 8.]

And our borderline accepted client — Client 8: score **55.84**, PD **33.50%**, expected loss **\$13,402**. He's accepted, but he's the most expensive client we say yes to.

[Pause for 2 seconds. Mention this business insight clearly.]

The point isn't just "yes or no" — the dashboard is already telling the bank **which** accepted clients are riskier and need extra attention.

[Click step 7 in the sidebar — Expected Loss.]

We're now on **Slide 7 — Expected Loss Dashboard**. This page rolls up the dollar impact.

[Point at the page subtitle showing the formula.]

The formula at the top: **Expected Loss = PD × LGD × Exposure**. Plain English: how likely they are to default, times how much we lose per dollar lent — **40%** — times the loan amount — **\$100,000**.

[Point at the three metric cards.]

For the **14 accepted clients**:

- **Total Expected Loss: \$129,329.63**
- **Average PD: 23.09%**
- **Accepted Exposure: \$1,400,000**

[Emphasize this result.]

This **\$129,329.63** is the centerpiece of our results — it matches **Table 16 of the report**. It's the dollar amount the bank should be ready to absorb on this \$1.4 million book, assuming a 60% recovery rate.

[Point at the “Expected Loss per Accepted Client” bar chart.]

The bar chart underneath shows the **per-client expected loss**. Taller bars mean **more capital at risk on that single client**.

[Point at the tallest bar — Client 8 at ~\$13,400.]

The tallest bar is — unsurprisingly — **Client 8, around \$13,400**. He's accepted because his score clears 48.20, but he's individually our largest loss exposure.

[Point at the shortest bars on the left — clients 4 and 5 at ~\$5,100.]

The shortest bars are clients 4 and 5 at about **\$5,100** each.

[Pause. Mention this business insight clearly.]

The same loan size produces a very different expected loss depending on the borrower. That's what justifies the next step — **charging different interest rates to different clients**.

[Click step 8 in the sidebar — Risk Pricing.]

We're now on **Slide 8 — Risk-Based Interest Rate Pricing**.

[Point at the formula in the subtitle.]

The formula: **Minimum interest rate = PD × LGD**. In simple terms, the rate has to at least cover the expected loss on that specific loan — otherwise the bank is lending at a guaranteed loss.

[Point at the bar chart — clients sorted by rate.]

Every bar is one client's minimum break-even rate, sorted from lowest to highest. **Green = accepted, red = rejected**.

[Point at the gold dashed “Avg” line.]

The gold dashed line is the **average minimum rate across the 14 accepted clients — 9.24%**.

[Point at the leftmost green bars.]

The safest clients — clients **4 and 5** — only need **5.11%** to break even. That's the cheapest credit the bank can responsibly offer.

[Point at the rightmost green bar.]

Our borderline client — **Client 8** — needs **13.40%**. Still acceptable, but priced much higher to reflect the risk.

[Point at the two summary boxes below.]

The two boxes below the chart sum it up:

- **Lowest accepted client rate: 5.11%**
- **Highest rejected client rate: 28.23%** — that's Client 1, the score-zero applicant. To break even on him, we'd need to charge **28%** interest. No bank would lend at that rate to a consumer — it's unrealistic and unethical. That's exactly why he's rejected.

[Pause for 2 seconds. Mention this business insight clearly.]

Risk-based pricing means: instead of charging everyone the same rate, the bank quotes a rate that **matches the risk of that specific person**. Safer clients pay less, riskier accepted clients pay more, and clearly-too-risky clients are not lent to at all.

[Pause.]

That's the full picture in dollars. My colleague will now bring everything together, show the final portfolio, and run a live test on the model.

Transition → “Let's see the consolidated results, the final approved portfolio, and a live demo of the scorecard in action.”

Speaker 6 — Model performance, final portfolio, live demo & conclusion

Time: ~4–5 minutes · **Pages:** Slide 9 — **Performance** (/performance), Slide 10 — **Final** (/final), Slide 11 — **Test** (/test)

[Click step 9 in the sidebar — Model Performance.]

We're now on **Slide 9 — Model Performance Summary**. This page is a one-screen recap of the whole project.

[Point at the “Validation Radar” chart on the left.]

The radar chart on the left visualizes our three validation metrics — **AUC**, **KS**, and **Accuracy Ratio** — all converted to percentages. The shape extends well beyond the centre, which means the model has consistent, balanced predictive power, not a single lucky number.

[Point at the “Project Outcomes” list on the right.]

The list on the right is the **executive summary**. Reading top to bottom:

- **500** historical observations
- **36%** historical default rate
- **AUC of 0.6775** on historical data
- **14 out of 20** portfolio clients accepted
- **70%** acceptance rate
- **23.09%** average PD on the accepted book

[Mention this business insight clearly.]

In other words — we moved from a historical default rate of **36%** down to an expected default rate of **23%** on the accepted book. That's the **value-add of using a credit scorecard**.

[Scroll down briefly — the accept/reject and risk band charts here are recap charts. Skim them.]

The two charts at the bottom — accepted vs rejected and risk band distribution — are just a visual recap of what we saw earlier. We'll go straight to the final view.

[Click step 10 in the sidebar — Final Decision.]

This is **Slide 10 — Final Portfolio Decision**.

[Point at the gradient banner at the top — “14 Clients · \$1.4M”.]

The big banner is the **final approved portfolio**:

- **14 clients approved**
- **\$1.4 million in exposure**
- **\$129,329.63 total expected loss**
- **Average PD 23.09%, average minimum interest rate 9.24%**
- **6 clients rejected**

[Point at the two columns below — Accepted on the left, Rejected on the right.]

The two columns underneath show every client by ID, with their score and PD. On the left, **14 accepted clients**, all sorted by score. On the right, **6 rejected clients** — all with scores below 48.

[Pause for 2 seconds. Look at the audience.]

This is the deliverable. If we were presenting this to a credit committee, this page is what they'd sign off on. Now let's prove the scorecard actually works **live**.

[Click step 11 in the sidebar — Live Client Test.]

This is **Slide 11 — Live Client Test**. Here you can pick **any one of the 20 applicants**, run the scorecard, and see the exact result the bank would get in production.

[Open the dropdown at the top — "Quick select (MATLAB portfolio)" — and pick "Client #4 — Score 100 (Accepted)".]

Let's start with our safest applicant — **Client 4**. As soon as we select him, the applicant profile fills in on the left: **age 50, income \$54,000, homeowner, employed**.

[Point at the "Scorecard point breakdown" table on the right.]

You can see his **point breakdown live** — Age **+24.87**, Income **+23.84**, Housing **+21.10**, Employment **+25.12**. They add up to **94.94**, capped at **100**.

[Click the teal "Run credit test" button. Wait for the animation.]

Now we click **Run credit test** — the dashboard simulates the MATLAB pipeline. Score gauge animates...

[Once result appears, point at the gauge and the four cards.]

...and the result: **Score 100, PD 12.79%, Accepted, Low Risk, Expected Loss \$5,114, Minimum Rate 5.11%**. **Exactly** the numbers from Table 15 of our report.

[Open the dropdown again, pick "Client #1 — Score 0.0 (Rejected)". Click Run credit test.]

For comparison, let's try **Client 1** — the worst applicant. Score **0**, PD **70.59%**, **Rejected, High Risk**, expected loss **\$28,234**, rate **28.23%**. Same scorecard, opposite outcome — that's the model doing exactly what it's supposed to do.

[Optional — say only if time allows: change Client 1's income to \$80,000 and re-run.]

We can even play "**what if?**": if we manually change Client 1's income from \$21,000 to \$80,000 and re-run, the score jumps and the decision flips — showing how sensitive credit risk is to income, which matches the IV we saw on Slide 2.

[Pause for 2 seconds. Mention this business insight clearly.]

In production, this is the exact tool a credit officer would use — type the applicant's profile, click one button, get a defensible decision in under a second.

Conclusion

[Step back from the screen. Speak slowly. Eye contact with the audience.]

To wrap up — in this project we built a **complete AI-in-FinTech credit risk system**:

1. We **learned from 500 past clients** using MATLAB's Credit Scorecard Toolbox.
2. We **built a scorecard** from four simple predictors — age, income, housing, employment — and validated it with **AUC 0.6775** and **KS 0.2892**.
3. We **applied it to 20 new applicants**, accepting **14** and rejecting **6** using the data-driven cutoff of **48.2032**.
4. We **quantified the risk in dollars** — **\$129,329.63** of expected loss on **\$1.4 million** of exposure.
5. We **priced every loan individually** — from **5.11%** for our safest clients to **13.40%** for the riskiest accepted ones.
6. And finally, we built **RiskLens** — this dashboard — so that every decision is **traceable, explainable, and live**.

[Brief pause.]

The project shows that the value of AI in finance isn't replacing the credit officer — it's giving them a transparent, consistent, and defensible tool. Every accept, every reject, every interest rate on this dashboard ties back to **the same four facts on a loan application**.

[Smile. Final eye contact.]

Thank you very much for your attention — we'd be happy to take your questions.

Q&A — short answers (any member can pick up)

Question	Suggested answer
Why train and validate on the same 500 clients?	Project brief specifies this approach. With only 500 records, splitting train/test reduces sample size below what's reliable.
Is AUC 0.6775 “good”?	For a 4-variable scorecard, yes — defensible academically. A production bank model uses 30–50 variables and aims for AUC > 0.75.
Where does 48.2032 come from?	It's the score that maximizes the KS statistic — i.e. the biggest separation between good and bad clients on the historical data.
What's the difference between WOE and Points?	WOE is a raw log-ratio per bin. Points are the scaled, rounded version of WOE × the regression coefficient, fitted to a 0–100 range.
Why is Client 8 accepted with a 13.40% rate?	His score (55.84) is above 48.20, so the model accepts him. The high rate just reflects that he's our riskiest accepted client.
What happens if the economy changes?	This is the main limitation — the model is trained on past data. In production, the bank would monitor PD vs realized defaults quarterly and recalibrate as needed.
Could this be improved with more data?	Absolutely. Adding credit-bureau scores, debt-to-income ratios, transaction history, and employment tenure would lift AUC significantly.
Where is the code?	matlab/FinTech_Scorecard_Project.m and the Final/ folder in the GitHub repository.

Speaker quick-reference (cheat card)

#	Speaker	Pages	Key numbers to mention
1	Intro & dashboard tour	Slide 1	500, 20, AUC 0.6775, 14 accepted, \$1.4M, \$129,329
2	Problem, data, methodology	Slide 1	36%, 320/180, 57% / 21%, \$100k loan, LGD 40%
3	WOE & Scorecard	Slides 2–3	IV: 0.2355 / 0.1879 / 0.1143 / 0.0417, points –8.68 to +28.91
4	Validation & Portfolio	Slides 4–5	AUC 0.6775, KS 0.2892, cutoff 48.2032, 5 / 9 / 6 risk bands
5	Decisions, EL, Pricing	Slides 6–8	14/6, \$129,329.63, \$1.4M, avg PD 23.09%, rates 5.11%–28.23%
6	Performance, Final, Live demo	Slides 9–11	70% acceptance, Client 4 → 100/12.79%/\$5,114, Client 1 → 0/70.59%/\$28,234

Rehearsal checklist

- ☐ Dashboard running and loaded on Slide 1 before audience walks in
 - ☐ Every speaker has clicked through their own slides once
 - ☐ Client Test page tested with both Client 4 and Client 1
 - ☐ Sidebar visible on the projection (not zoomed out)
 - ☐ Backup report PDF open on a second device
 - ☐ Internet not required — `npm run dev` runs locally
-

Script aligned with `Final/fintech (1).pdf` (Tables 13 / 15 / 16), `matlab/FinTech_Scorecard_Project.m`, and the RiskLens dashboard (`fintech-dashboard/src/config/presentationNav.ts` , steps 1–11).